Rohan Mehra

$ role: Full-Stack Systems & AI Infrastructure Engineer

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LinkedIn
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Portfolio

// TECHNICAL ARSENAL

`Rust` `Python` `Go` `TypeScript` `PostgreSQL` `Redis` `Kafka` `Docker`

Practices: Agile Collaboration · Code Reviewer

// PROJECTS & OPEN SOURCE

StreamVortex: Real-Time Telemetry & Log Aggregator

`Creator & Lead Developer`

High-volume stream processing engine designed for real-time log parsing.
Processes billions of incoming events per day with distributed partitioned consumers.
Features custom query syntax and instant sub-second histogram visualizations.

- Engineered zero-copy memory ring buffer receiving 500,000 UDP metric packets per second.
- Wrote custom ClickHouse ingestion pipeline reducing storage footprint by 64% using LZ4 block compression.
- Built WebSocket dashboard rendering 60 FPS live charting for cluster CPU, RAM, and disk utilization.
- Implemented automatic anomaly detection triggering PagerDuty alerts on unusual latency spikes.
- Authored end-to-end integration test harness simulating chaos monkey network dropouts.

`[Rust | Kafka | ClickHouse | React | TypeScript]`

Live Demo · Source Code

NeuralServe: Low-Latency Model Inference Proxy

`Backend Architect`

Distributed model inference proxy supporting dynamic request batching and GPU memory scheduling.
Provides automatic model hot-reloading without dropping active client connections.

- Decreased P99 inference latency from 140ms to 24ms by implementing dynamic micro-batching.
- Integrated token bucket rate limiter preventing upstream GPU out-of-memory crash cascades.
- Built health check supervisor automatically failing over between primary and standby worker nodes.
- Supported streaming server-sent events (SSE) for LLM response token generation.

`[Python | FastAPI | ONNX Runtime | Triton | Redis]`

Live Demo · Source Code

// WORK EXPERIENCE

Backend Engineering Intern · Zepto Delivery Systems

`Jan 2024 – Jun 2024`

Enhanced rider assignment routing service handling rapid order dispatches during peak hours.
Collaborated directly with product and operations teams to optimize dispatch dispatch efficiency.

- Refactored geospatial Redis geohash queries decreasing rider matching time by 38%.
- Built automated alerting pipeline tracking order drop-offs and failed delivery notifications.
- Mentored 3 junior summer interns on Golang testing methodologies and code linting standards.

// EDUCATION

College of Engineering Pune (COEP)
B.Tech in Computer Engineering in Computer Engineering

2021 – 2025
GPA: 8.85 / 10.0